

Name: _____ Date: _____

Period: _____

Parametrization Check

The ellipse $\frac{(x+1)^2}{9} + \frac{(y-2)^2}{16} = 1$ is implicitly defined.

1. Identify h , k , a , and b from the equation. Then write the parametric equations that trace the ellipse counterclockwise for $0 \leq t \leq 2\pi$.

$h =$ _____ $k =$ _____ $a =$ _____ $b =$ _____

$x(t) =$ _____ $y(t) =$ _____

2. Using your parametrization from Question 1, find the coordinates of the topmost and bottommost points on the ellipse. Show how you use $y(t)$ to find them.
